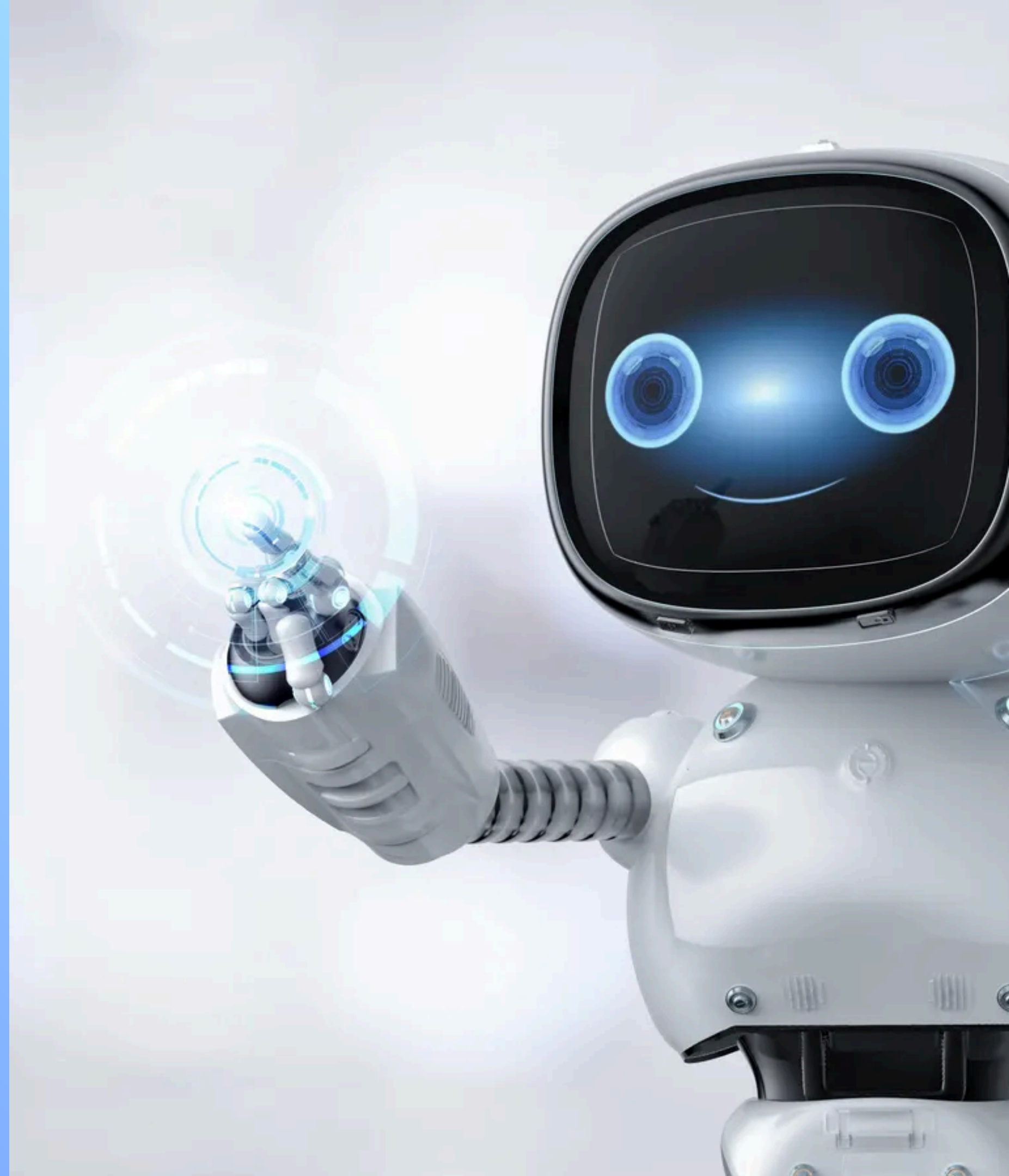


WALL PAINTING ROBOT

GROUP 01

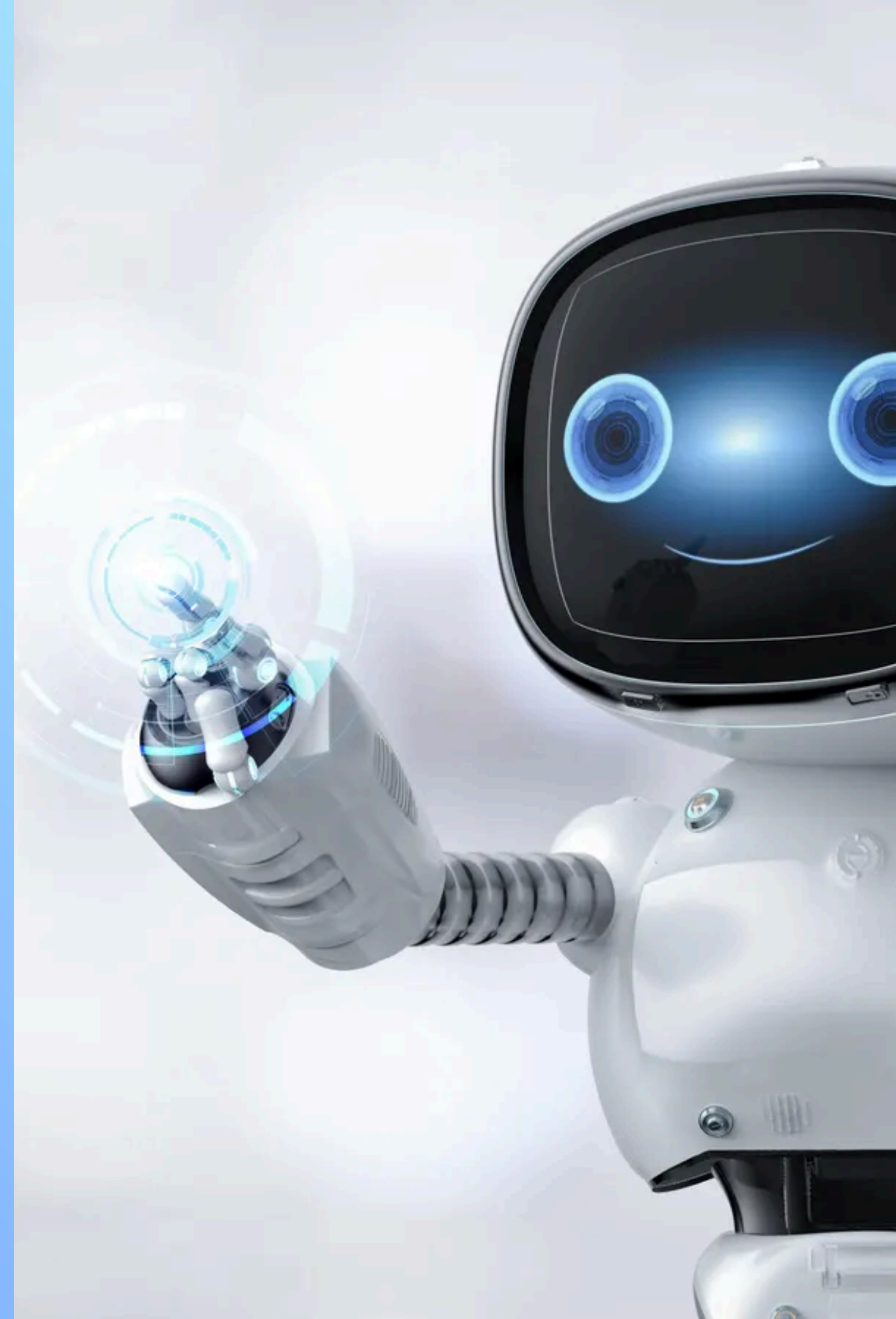
Dilusha
Daham
Danuka
Shenal
Nethmini

Anuka
Sanju
Nethmina
Kathurshan
Thivmal



Introduction

Our project presents an automated wall painting system that improves efficiency and safety by reducing labor, time, and exposure to hazards. It ensures uniform painting with minimal human involvement.



SYSTEM OVERVIEW

1



Navigation System

2



Color Detection

3



Painting Mechanism

4

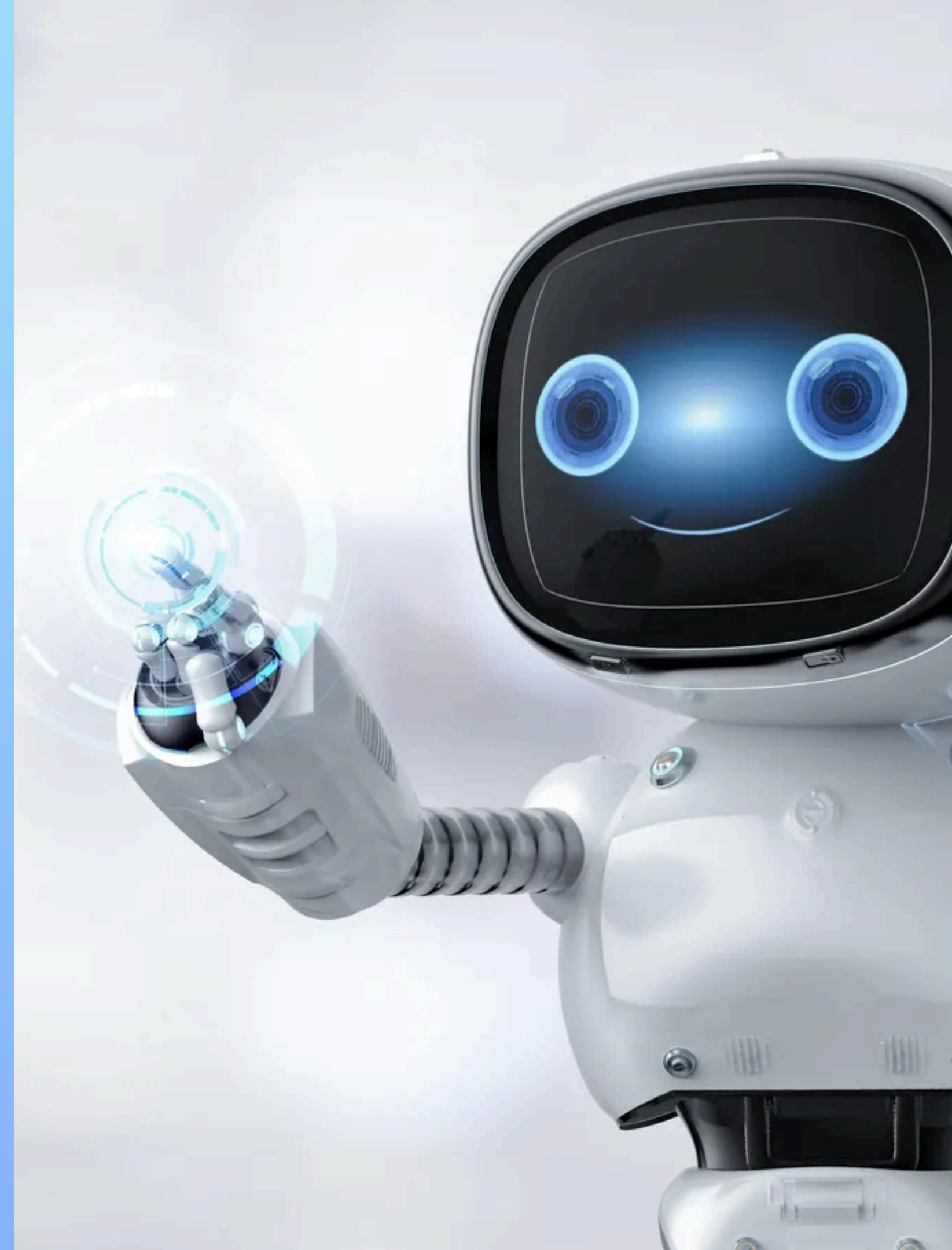


Paint Monitoring

5



Mobile App Development



Hardware Components Used

1. Mobility System

- Meachum Wheels - Enables omnidirectional movement, allowing the robot to move in any direction (forward, backward, sideways, and rotate).
- Gear Motors - Provide the necessary torque for movement.
- L298N Motor Driver - Used to control the gear motors' direction and speed.

2. Position & Motion Sensing

- HC-020K Rotary Encoders - Measure wheel rotation to track movement and distance traveled with precision.

3. Distance & Obstacle Sensing

- Ultrasonic Sensors - Used for obstacle detection and wall distance measurement to ensure accurate painting alignment.

4. Color Detection

- TCS3200 Color Sensor - Detects wall or surface color to adapt painting or avoid repainting already colored areas.

7. Control & User Interface

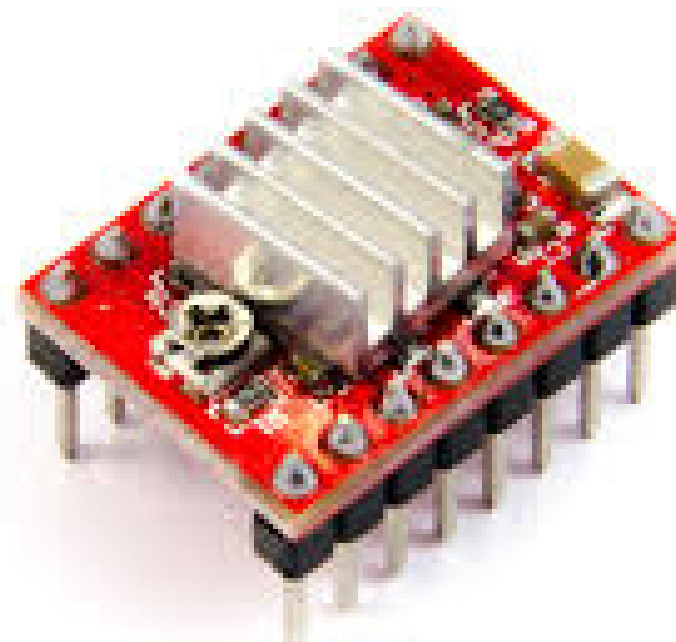
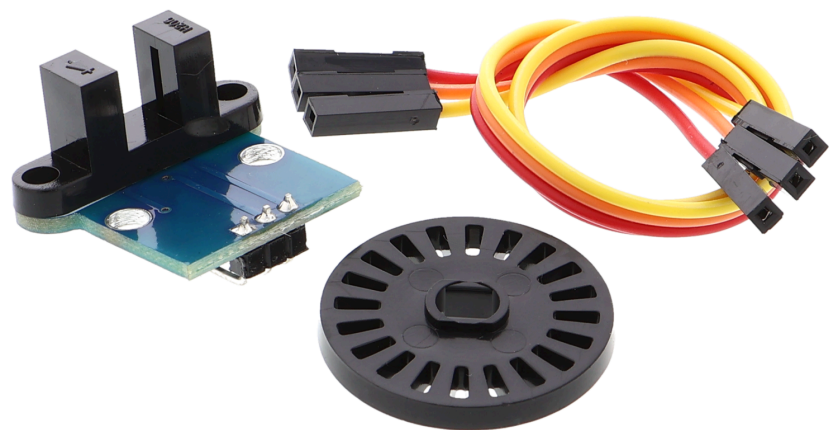
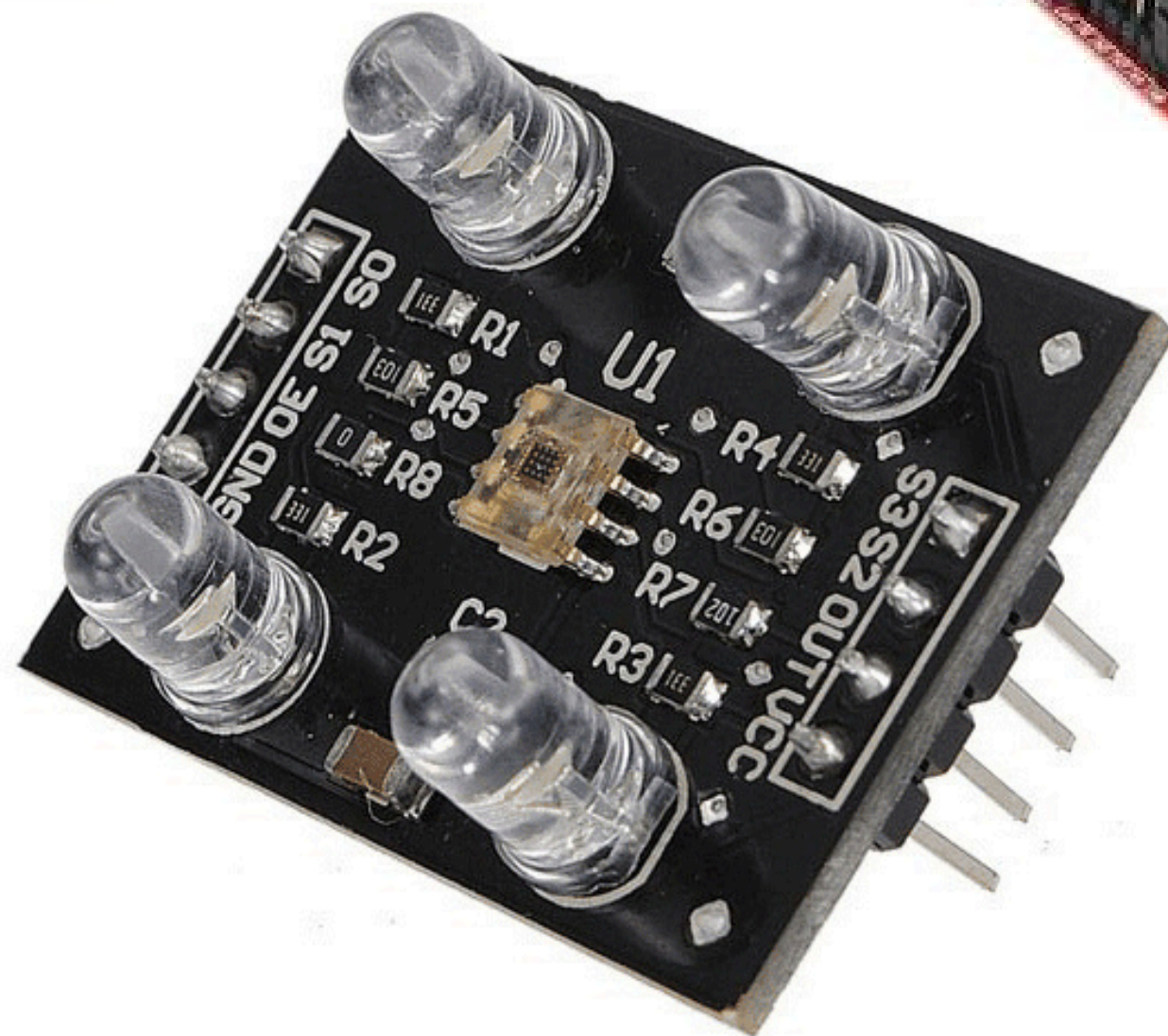
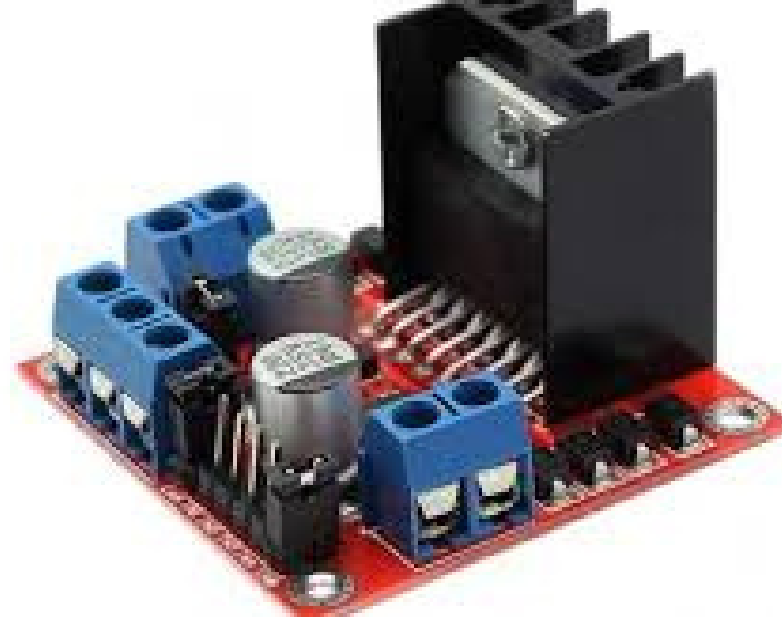
- Custom Mobile App (MIT App Inventor) - Provides a user-friendly interface to control the robot's movement, painting actions, and mode switching.

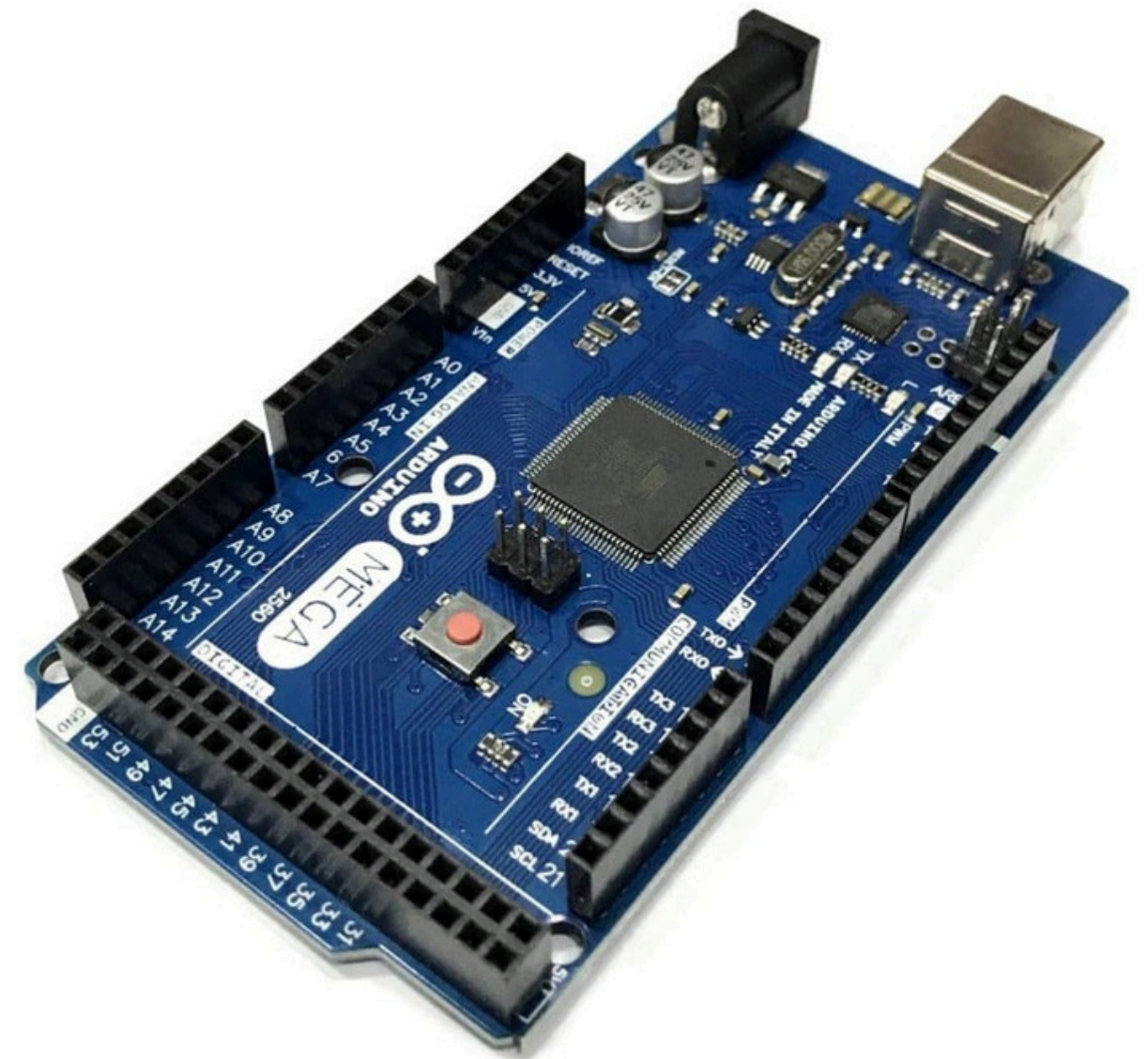
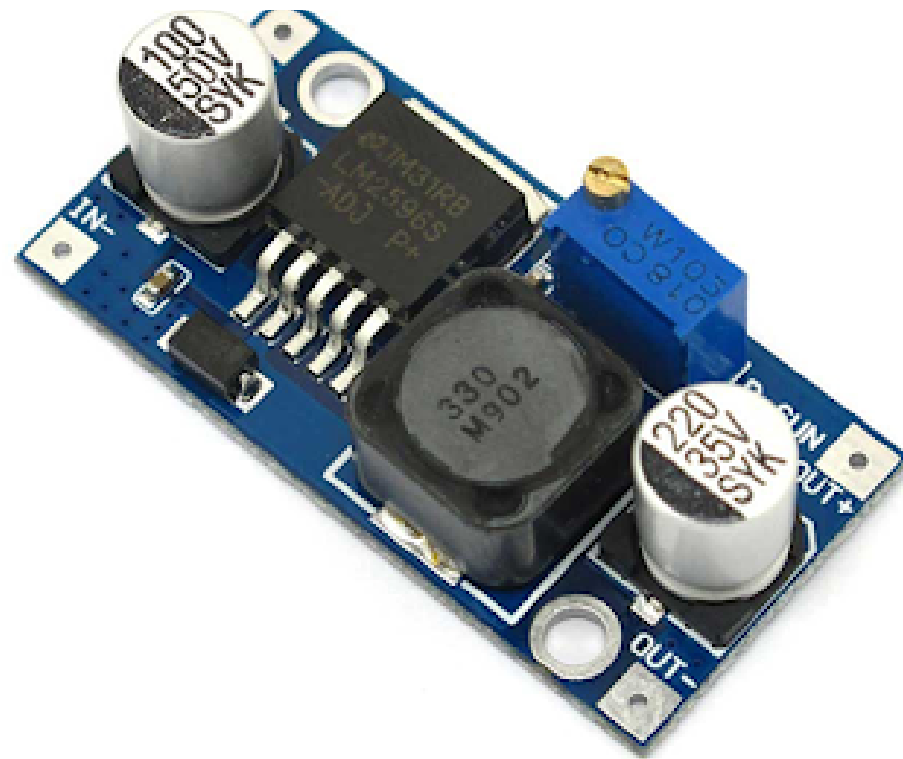
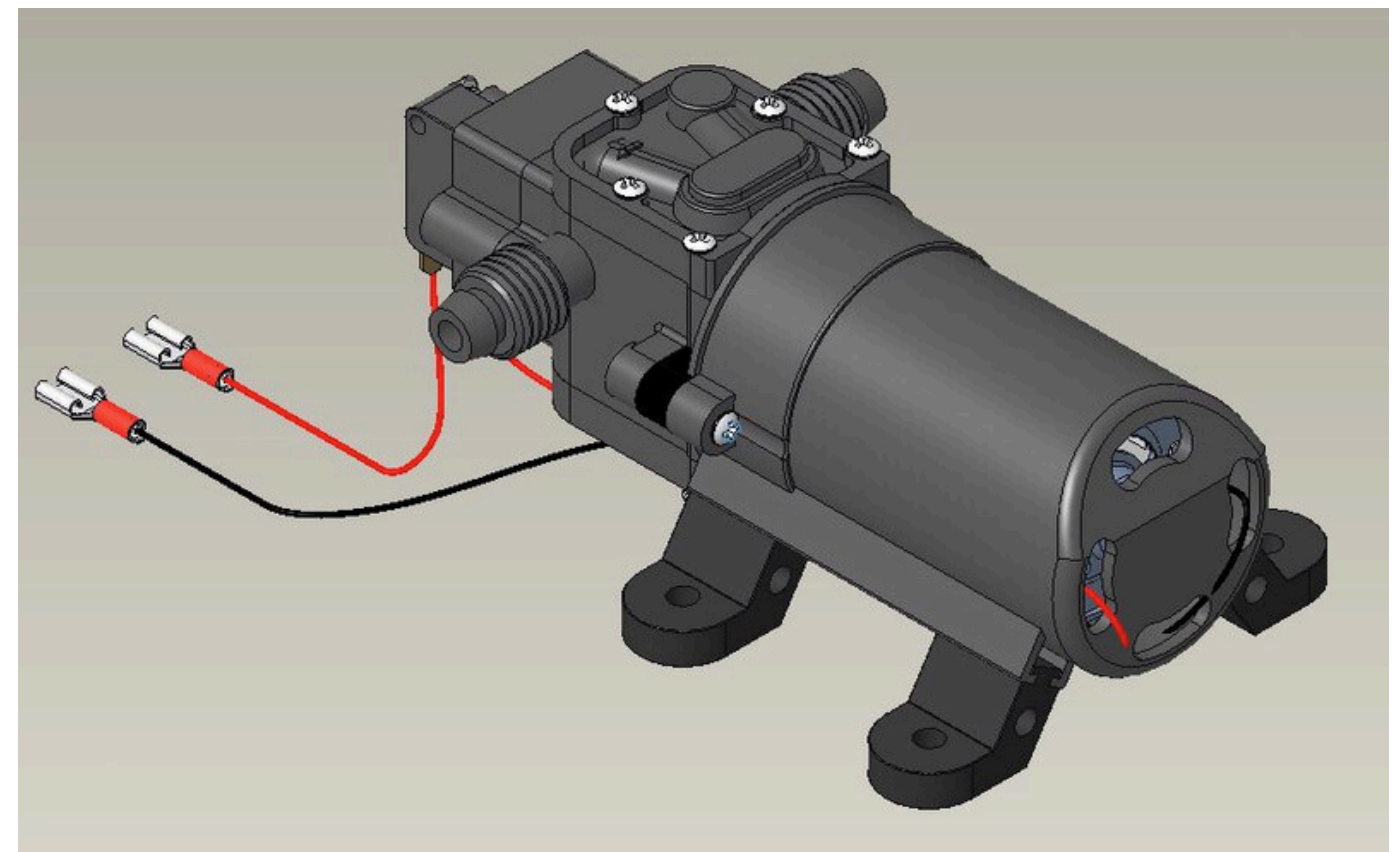
5. Painting Mechanism

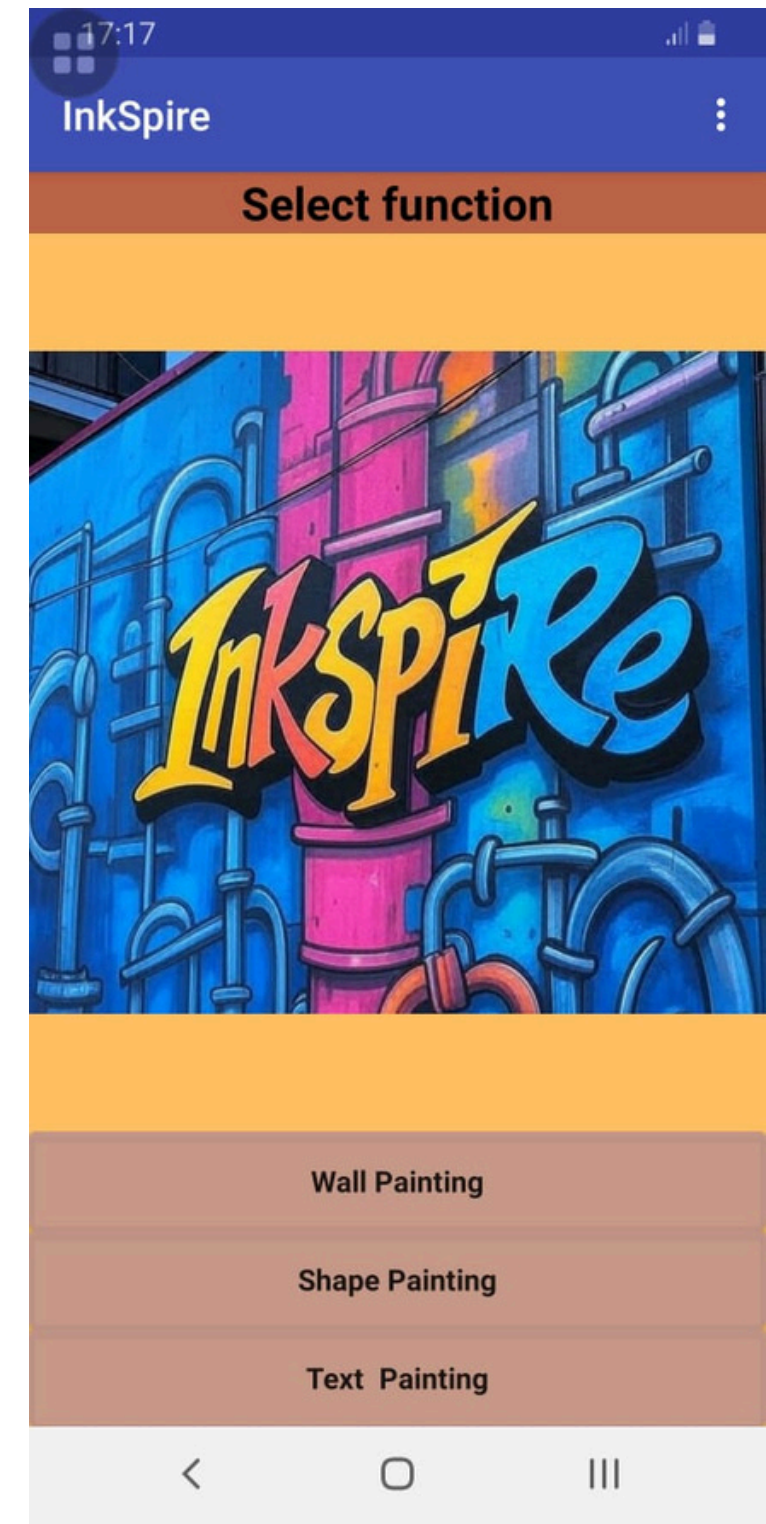
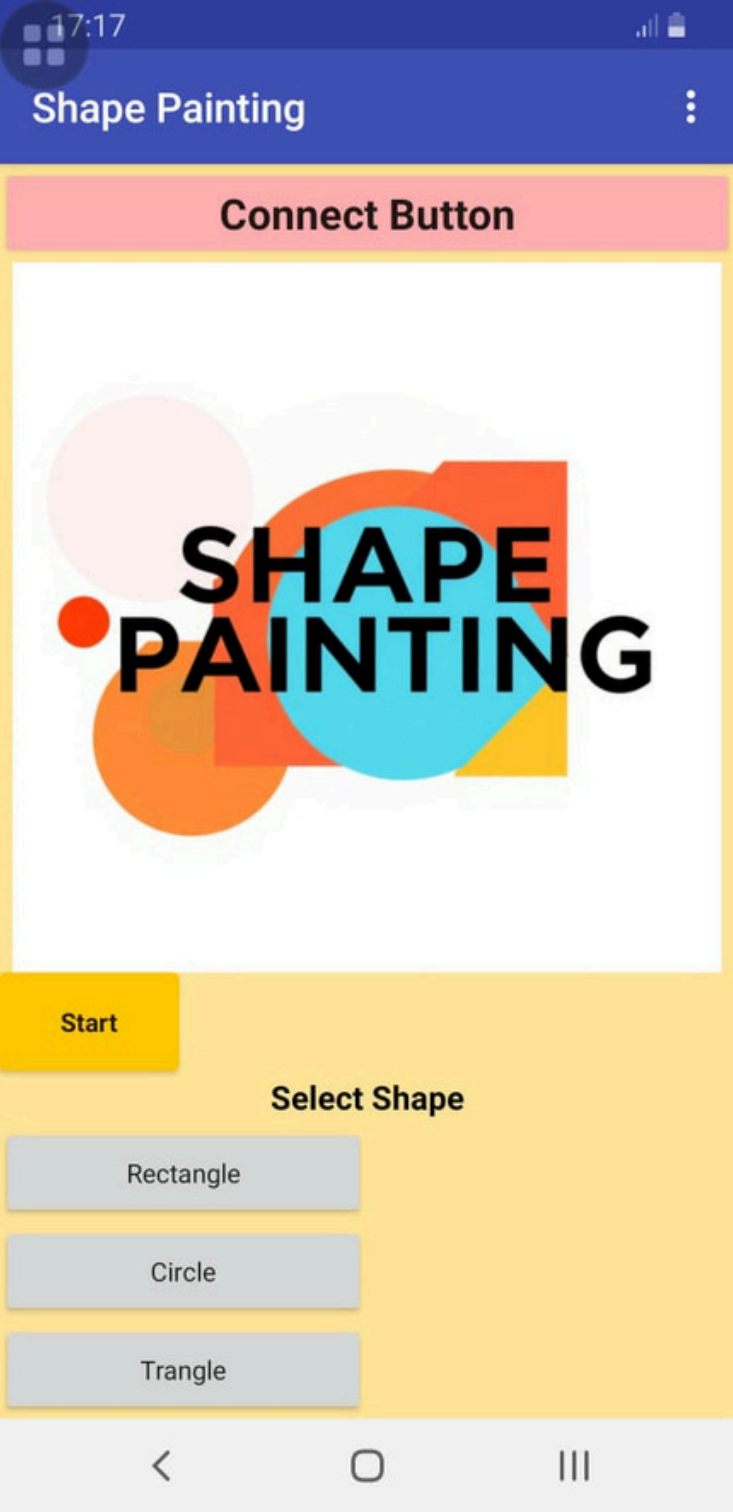
- Mini Water Pump + Nozzle - Delivers paint in a controlled manner over the wall surface.

6. Actuation System

- NEMA 17 Stepper Motor - Provides precise linear movement for painting mechanisms (e.g., vertical arm lift).
- A4988 Stepper Driver Module - Controls the NEMA 17 motor with high precision and reliability.







when ListPicker1 .BeforePicking
do set ListPicker1 . Elements to BluetoothClient1 . AddressesAndNames

when ListPicker1 .AfterPicking
do set ListPicker1 . Selection to call BluetoothClient1 .Connect
address ListPicker1 . Selection
set ListPicker1 . Text to " Connected "

when Start .Click
do if BluetoothClient1 . IsConnected
then call BluetoothClient1 .SendText
text " D "

when Send .Click
do if BluetoothClient1 . IsConnected
then call BluetoothClient1 .SendText
text TextBox1 . Text

when Screen2 .BackPressed
do close screen

when ListPicker1 .BeforePicking
do set ListPicker1 . Elements to BluetoothClient1 . AddressesAndNames

when ListPicker1 .AfterPicking
do set ListPicker1 . Selection to call BluetoothClient1 .Connect
address ListPicker1 . Selection
set ListPicker1 . Text to " Connected "

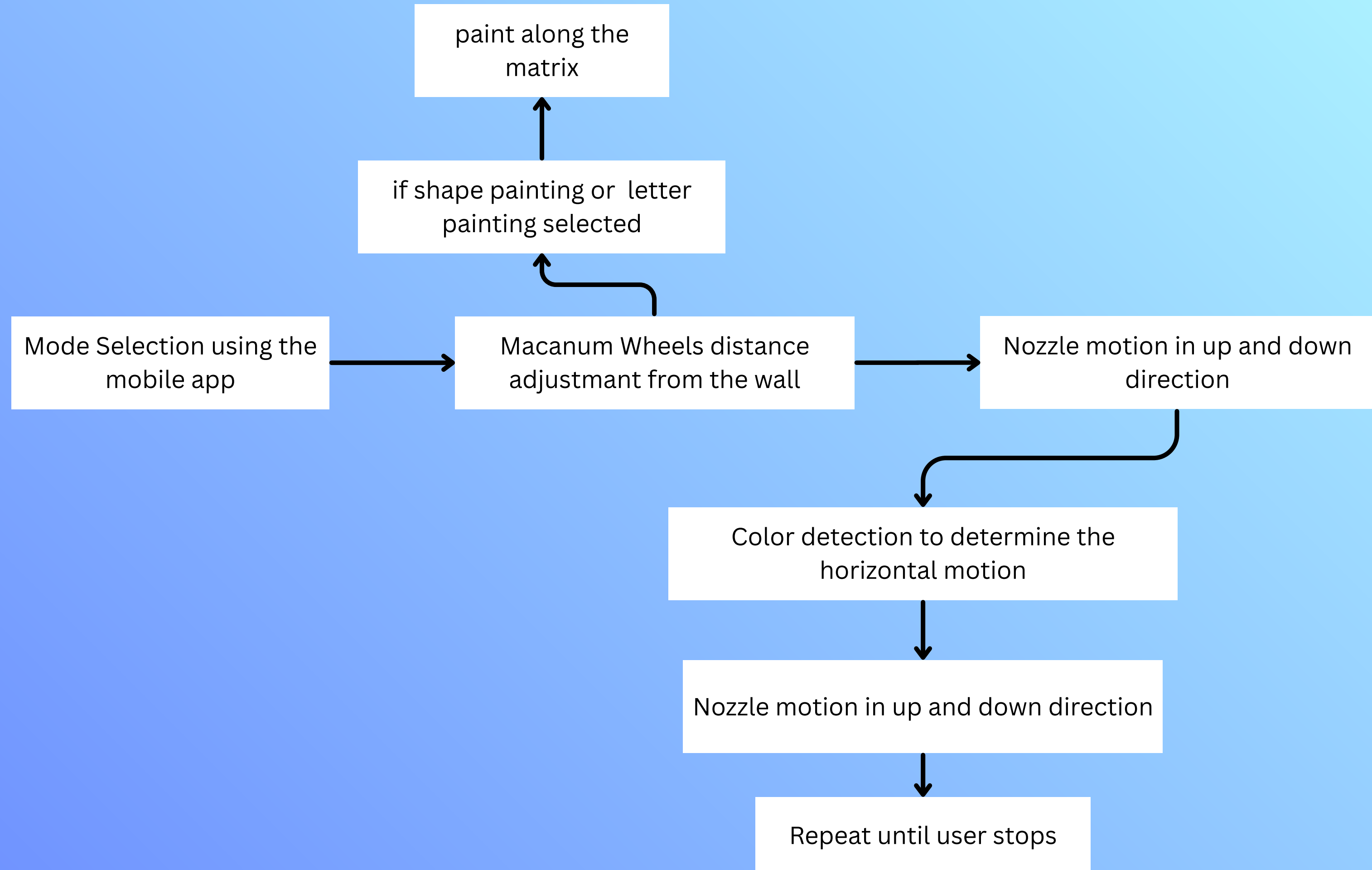
when start .Click
do if BluetoothClient1 . IsConnected
then call BluetoothClient1 .SendText
text " E "

when rectangle .Click
do if BluetoothClient1 . IsConnected
then call BluetoothClient1 .SendText
text " R "

when circle .Click
do if BluetoothClient1 . IsConnected
then call BluetoothClient1 .SendText
text " O "

when trangle .Click
do if BluetoothClient1 . IsConnected
then call BluetoothClient1 .SendText
text " T "

when Screen4 .BackPressed
do close screen



Thank You!